

Aspect-based sentiment analysis through zero-shot text classification and impact-asymmetry analysis

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ABSTRACT

This study proposes innovative methods integrating aspect-based sentiment analysis (ABSA) using zero-shot text classification (ZSTC) into the impact-asymmetry analysis. Although ABSA is essential to enrich a thorough understanding of customer behavior, applying it to hospitality and tourism management is challenging because matching sentiment valence with an aspect category is complicated, and annotated datasets are scarce. The ZSTC suggested by this study leverages pre-trained large language models and addresses these challenges in hospitality and tourism management contexts. Furthermore, impact-asymmetry analysis based on the Kano model investigates the asymmetric relationships between the identified aspects and customer satisfaction. It classifies them into satisfiers, dissatisfiers, or hybrids and prioritizes them to support managerial decision-making. A case study analyzing TripAdvisor reviews during the COVID-19 pandemic is provided to demonstrate the usefulness of the proposed methodology. This study's findings offer theoretical and practical contributions to the academic and business sectors.

1. Introduction

Digitalized textual data has proliferated with the advent of the internet. Companies analyze user-generated content on social media or online review sites as a new source of understanding the perceptions and attitudes of customers (Dwivedi et al., 2021). Fast-growing natural language processing (NLP) skills have facilitated this work. Among NLP tasks, sentiment analysis (SA), or opinion mining, is widely used in this kind of analysis. SA has advantages over conventional star ratings given along with online reviews. As a tool for understanding customer behavior, star ratings are often biased, failing to reflect the complexity of customers' experiential perceptions and emotions on products or services (Al-Natour and Turetken, 2020). As noted by Gaspar et al. (2016), not only positive but also negative emotions can be expressed in a review, though a rating is high. As such, star ratings alone restrict a deeper understanding of customers' minds.

SA can be conducted at a review, sentence, or aspect level (Yadav and Roychoudhury, 2019). Despite its usefulness for managerial decision-making, an aggregated sentiment value that presents a single representative sentiment at a review level has limitations in enhancing the understanding of customer behavior because it does not contain rich and compound information. Moreover, if managers want to know what and how much they need to improve in customer service, the aggregated

sentiment will not be of much help. Even if they are aware of the factors of interest, without insights into customer perceptions of each factor, they would be uncertain about where to prioritize their efforts. In this regard, certain online review sites, such as TripAdvisor, present star ratings for pre-defined aspects. However, these aspects are constrained, and the ratings provided by reviewers are often sparse (Luo and Tang, 2019).

Hence, aspect-based sentiment analysis (ABSA) has continuously attracted researchers' interest. ABSA is a fine-grained model of SA that provides sentiment information regarding specific features of products and services considered as an aspect (Schouten et al., 2018; Tubishat et al., 2018). By shifting the focus from an entire document to specific aspects, ABSA captures segmented sentiments depending on the various aspects of products or services (Zhang et al., 2023). In the context of hospitality and tourism, online reviews provide enriched information on customers' perceptions and attitudes regarding their experiences. However, hospitality services present unique analytical challenges due to their multidimensional nature. ABSA specifically addresses these challenges by enabling the exploration of relationships between different service aspects, which can theoretically contribute to hospitality management research. This approach allows hospitality managers to gain a more nuanced understanding of customer evaluations of diverse factors and make prioritized improvement decisions based on

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their relative impact on satisfaction.

However, whereas discussion about ABSA led by computational linguistics has mainly focused on enhancing the accuracy of sentiment classification, few studies have explored the adaptation of ABSA to a hospitality and tourism management context. Indeed, ABSA focused on the methodology itself is less helpful in uncovering why and how problems latent in sentiments based on aspects (Kar and Dwivedi, 2020). Moreover, ABSA requires a large volume of pre-annotated datasets since it has been considered a classification problem in machine learning. Regarding the dataset with annotated labels of aspects, studies have relied on the limited availability of the aspect-sentiment-labeled dataset, such as SemEval-2016 (Pontiki et al., 2016), mainly in the laptop or restaurant domain. It becomes problematic when researchers or managers apply ABSA to other domains where the labeled dataset does not exist. For instance, new phenomena such as COVID-19 that do not have labeled samples, where the annotation is laborious and challenging (Pourpanah et al., 2023), cannot be investigated by this approach.

Therefore, this study attempts to bridge a critical research gap by advancing ABSA and combining it with a theoretical framework in the hospitality management. Unlike computational research that has primarily focused on technical accuracy improvements, this work addresses the specific challenges faced by hospitality researchers and practitioners: the need to analyze customer sentiment without extensive labeled datasets, and the requirement to transform this analysis into actionable service improvement priorities. It specifically addresses the following research questions.

- **RQ1.** How can aspect-based sentiment polarities be effectively detected from online reviews in the hospitality and tourism field “when annotated data are limited”?
- **RQ2.** How can aspect categories and sentiment valence be utilized more effectively for further quantitative analysis?

For the RQ2, this study induces the Kano model (Kano et al., 1984) that notes the different features of products have asymmetric effects on customer satisfaction. It proposes three categories classifying those features: attractive, must-be, and one-dimensional quality features (Bigorra et al., 2019). Attractive features are related to excitement by fulfilling expectations, must-be features are related to mere customer dissatisfaction by not fulfilling expectations, and one-dimensional features can be related to both. The Kano model enables a more in-depth perspective on the complex relationships between the outcomes of ABSA and customer satisfaction, which contributes to a better understanding of customer behavior as well as decision-making.

Maintaining this perspective, this study proposes a framework encompassing ABSA using zero-shot learning and an asymmetric analysis of the aspect categories toward satisfaction. Zero-shot learning classifies instances into labels that have not been trained previously, which has the potential for classification tasks in NLP (Yin et al., 2019). In this study, a novel approach identifies the aspect category and the sentiment valence without the term-category pair or any annotated label. Furthermore, compared to the conventional linear relationship analysis, asymmetric analysis unveils dynamic relationships of identified aspects toward satisfaction. Details are explained in the proposed methodology section, and a case study adapts the proposed framework to online hotel reviews under the COVID-19 pandemic in which labeled samples are rare.

2. Related work

2.1. Research on the aspect-based sentiment analysis (ABSA)

Sentiment analysis (SA) refers to evaluating the emotional tone in written text to comprehend the affective states expressed in the content (Raman et al., 2022). Generally, SA identifies the sentiment polarity in the text at a document, sentence, or aspect level, using a dictionary,

machine learning, or deep learning-based approach. The dictionary-based approach employs a pre-defined dictionary or lexicon, such as SentiWordNet (Baccianella et al., 2010), with a pair of words and sentiment scores. Machine learning and deep learning-based approaches generally use annotated data and establish a classifier through supervised learning, which can then be used to assess the sentiment of a given document (Jeong et al., 2019).

Recently, attention to ABSA has grown because customers' affection for a product or a service is often based on a specific feature rather than an overall assessment (Venugopalan and Gupta, 2022). The aspect elements consist of an aspect term and an aspect category (Zhang et al., 2023). While capturing aspect terms in the text is straightforward, they are often too intricate and detailed for direct use. In contrast, the aspect category, being more concise, topical, thematic, and perceptual, is more advantageous for exploring ABSA problems. Consequently, identifying aspect categories derived from explicit aspect terms becomes critical for fulfilling the goals of ABSA. For instance, given the sentence “The room with a sofa bed came with no pillows,” determining “room amenities” from the fine-grained aspect terms, such as “sofa bed” and “pillow” as an aspect of this sentence is demanding but more effective. As noted, one of the critical issues of the ABSA is that annotated data is limited, which means there is the least standard foundation to which the identification of aspects refers.

In the unsupervised learning approach, this issue is not problematic because it does not need annotated data for label reference. Instead, categories are determined from text. For instance, topics generated by topic modeling techniques, such as LDA, can be treated as aspects of reviews. Then, keywords aligned with each topic are used to assess different aspects present in the text when they match the words used. Luo and Tang (2019) followed this approach and evaluated sentiment polarity by linking terms with external sentiment lexicons. The disadvantages of this approach lie in the fact that aspects are constrained to topics generated by LDA, which hinders conducting SA based on a more comprehensive range of aspects. Moreover, this approach is unavailable for a small number of reviews due to the limitation of the LDA technique. On the other hand, Yadav and Roychoudhury (2019) and Hu et al. (2021) captured the nouns that most frequently occur, using POS tagging, and treated them as aspects. In this case, the aspects become granular, and another process for applying those aspect terms to aspect categories is necessary.

The supervised learning approach assumes that annotated data with labels are available. This classification requires a training process with a well-annotated dataset, which is an obstacle to ABSA. For instance, Li et al. (2023) employed BERT to perform ABSA in a restaurant context. Although BERT is generally more reliable than the previous lexicon-based methods, it is designed for fine-tuned tasks and requires a sufficient size of labeled datasets to perform effectively (Widmann and Wich, 2023). At least, despite being pre-trained, BERT cannot be applied to other contexts or used to investigate certain new aspects arising from novel research questions.

2.2. Zero-shot text classification (ZSTC) based on natural language inference

Zero-shot learning originated from computer vision to classify new images into classes that are not trained (Lampert et al., 2014). Later, it has been increasingly used in NLP tasks, such as language translation, entity extraction, classification of new semantic classes, and relation extraction (Wang et al., 2019). Zero-shot text classification (ZSTC) refers to classifying text into candidate classes using zero-shot learning to predict labels that are not included in the training set. Whereas common machine learning or deep learning struggles with a shortage of training samples, ZSTC can deal with emerging classes that have not appeared in previous datasets (Chen et al., 2021). ZSTC can classify multiple aspects that may be contained even within a single sentence. For example, Yin et al. (2019) identified ‘topic,’ ‘emotion,’ and ‘situation’ aspects,

respectively, included in one sentence.

Early ZSTC attempted to train a large corpus of sentences using a single neural network and transfer a developed model into a new dataset to predict unseen classes (Pushp and Srivastava, 2018). However, the recent development of pre-trained large language models (LLMs) has substantially improved the performance of ABSA problems by not only introducing enhanced embeddings but also providing feasible solutions for diverse downstream ABSA tasks (Zhang et al., 2023). While LLMs are pre-trained language models that can be used in many different types of downstream tasks, they are generally fine-tuned for the tasks. Accordingly, gaps exist between pre-trained and fine-tuned models from the transfer learning perspectives (Han et al., 2022). Moreover, in the ZSTC problem, transferring knowledge from seen to unseen classes can be challenging because shared patterns or characteristics between these classes are scarce (Ye et al., 2020). Hence, a semantic tool should help bridge pre-trained knowledge and classified categories to improve the classification quality beyond generalization.

Among many attempts to mitigate the gaps, prompt-based approaches in using LLMs have recently attracted attention. This approach originated from the task of natural language inference (NLI) (Dagan et al., 2006) that aims to predict the relationship between two sentences, such as entailment, contradiction, or neutral (Sainz and Rigau, 2021). For example, ZSTC can be improved by providing appropriate prompts using a next-sentence prediction model (Zhao et al., 2022). With the advancement of LLMs, prompt engineering-based approaches have been continuously progressing. Despite significant progress in ZSTC in computational linguistics, including developments in prompt engineering, the field of hospitality and tourism management still lags behind in incorporating this technology into its research.

Moreover, it is crucial to distinguish LLMs used for ZSTC approaches from recent generative LLMs (e.g., ChatGPT). Most generative LLMs are fundamentally designed for text generation through autoregressive architectures optimized for next-token prediction (Brown et al., 2020). In contrast, ZSTC methodology requires NLI capabilities that evaluate entailment relationships between premises and hypotheses—a structured logical reasoning task distinct from text generation (Yin et al., 2019). NLI-based models are specifically trained to compute entailment probabilities for systematic classification. They produce deterministic outputs essential for reproducible academic research, which generative models cannot guarantee due to their stochastic nature.

2.3. KANO model and impact-asymmetry (IA) analysis

Recent works have criticized the assumption that various customer factors have linear relationships with customer satisfaction (Bi et al., 2019). Particularly, Kano et al. (1984) extend Herzberg's (1968) two-factor theory and propose a model consisting of three factors impacting customer satisfaction differently, such as attractive, one-dimensional, and must-be quality features. Attractive quality features relate to delight or excitement to customers. When they are absent or not fulfilled, it does not lead to customer dissatisfaction because customers do not have high expectations of them. However, if they are well offered to customers, it greatly impacts positive customer satisfaction. One-dimensional quality features refer to features with a conventional linear relation with customer satisfaction. When they are absent or not fulfilled as expected, customers will be dissatisfied, but if they are fulfilled, customers will be satisfied. On the contrary, must-be quality features do not influence customer satisfaction, even though they are performed well. However, their poor performance leads to customer dissatisfaction.

This perspective, despite adjustment and variation, has been applied to various works, such as quality function development (Liu et al., 2022), attribute performance in hotel services (Bi et al., 2020), selecting improvement options for multichannel retailers (Baier and Rese, 2020), or security perceptions on smartphones (Xiao, 2021). Unlike the

conventional Kano model, which relies on customer survey data, recent studies (e.g., Bi et al., 2020; Bigorra et al., 2019) actively take advantage of online textual reviews to capture product features aligned with customer satisfaction.

Meanwhile, the Kano model has been extended by integrating various impact-asymmetry (IA) analysis methods. For instance, penalty-reward contrast analysis (Brandt, 1987) has been employed in examining the nonlinear impact of aspects on overall satisfaction (Mikulić and Prebežac, 2008). The penalty-reward contrast analysis uses a multiple regression equation, where overall satisfaction plays a role as a dependent variable, using penalty or reward factors of features as independent variables. This approach identifies the non-linearity of aspects by distinguishing penalty factors from negative factors and can categorize aspects among the features in the Kano model (Pico and Tontini, 2018). However, the generic penalty-reward contrast analysis does not reveal the prioritized significance of those aspects.

Additionally, the method of importance-performance analysis (Martilla and James, 1977), which was developed as a distinctive analytical approach, is frequently combined with the Kano model (Hu et al., 2022; Pico and Tontini, 2018). For example, the different importances of aspects were measured by the coefficients of aspects in the regression equation (Bi et al., 2020; Xiao, 2021). This combination aids in closely examining the priorities of various aspects to enhance, considering their asymmetric effects on overall satisfaction. It is worth noting that three factors in the Kano model have varied in their names, depending on the different perspectives, though the core concept remains consistent. For example, attractive, one-dimensional, must-be features in the Kano model are called excitement, performance, and basic factors, respectively in Bi et al. (2020), while delighters, satisfiers, hybrids, dissatisfiers, and frustrators were referred to from the perspective of customer satisfaction (Lee et al., 2020).

3. Proposed methodology

Based on reviewing the literature and finding the limitations of the present ABSA research mentioned above, this study offers a methodology incorporating ABSA using ZSTC into Kano-model-based IA analysis. Fig. 1 describes the entire process of the analysis conducted in this study.

3.1. Data preparation

During the web scraping process, because it is impractical to selectively harvest user-generated content of specific interest from social media sites, researchers typically gather a broad range of reviews initially. This study applied Word2vec (Mikolov et al., 2013) to identify particularly focused reviews for a case study, as described later in the case study section.

In conventional NLP research, texts are normally broken into semantic unit elements. However, this task (so-called tokenization) is not necessary in the proposed methods because ZSTC using pre-trained LLMs works on a contextual basis and does not need any further fine-tuning. Likewise, the proposed model does not require text pre-processing, such as lowercasing, removing stop words, and stemming or lemmatization, leaving text as it is. Instead, constructing auxiliary sentences is recommended to increase the clarity of each sentence containing pronouns (Murtadha et al., 2022). Dealing with the sentence, the demonstrative pronoun can be replaced by named entities (Chauhan et al., 2022). By doing so, aspects and sentiments implied in each sentence can be revealed in the context. For instance, sentences, "My sister lives in Houston. She is quite happy in that city." are modified to "My sister lives in Houston. My sister is quite happy in Houston." using the 'NeuralCoref' and 'spaCy' libraries in Python. After the auxiliary sentences are constructed, each sentence is split and kept separately to obtain ABSA polarities based on the sentence.

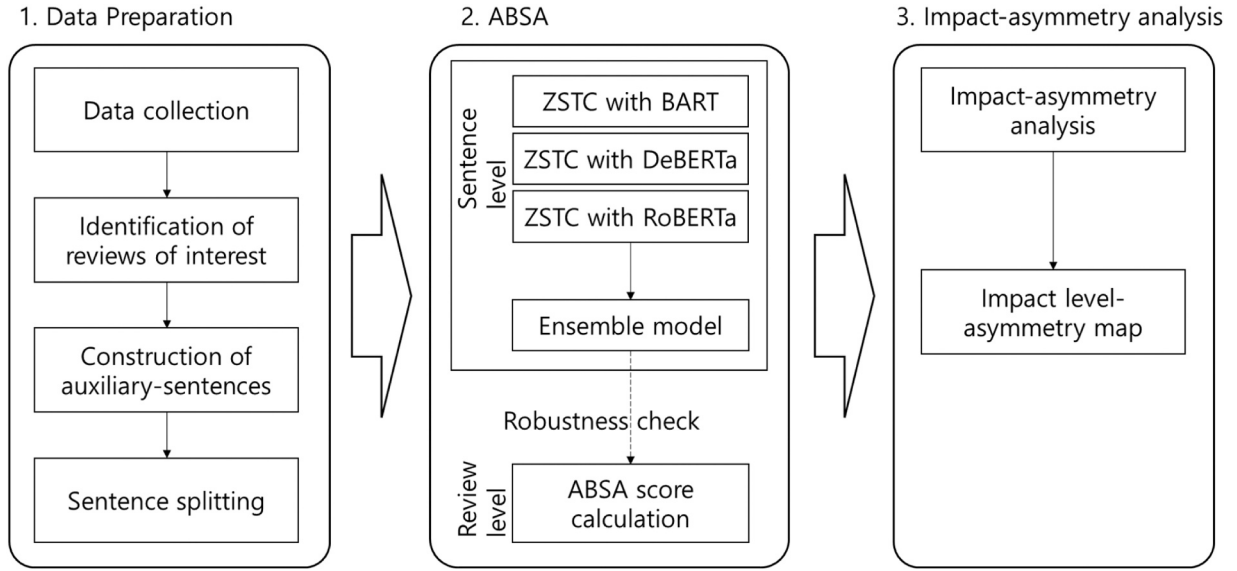


Fig. 1. The proposed methodology.

3.2. Aspect-based sentiment analysis (ABSA) using zero-shot text classification (ZSTC)

Annotated data are largely unavailable in most domains. Pre-trained zero-shot learning LLMs available on the Hugging Face platform (<http://huggingface.co/>) can assign labels to unannotated text using any user-defined categories, which can be applied for aspect categorization, emotion detection, relation extraction, and more kinds of classification problems (Yin et al., 2019). As ABSA has two distinctive classification problems (i.e., aspect classification and sentiment classification), researchers can design different candidate sets of classes for an aspect and a sentiment, respectively. Then, ZSTC is conducted with an instruction-tuned pipeline consisting of hypothesis and candidate options to improve performance based on the NLI task (Wei et al., 2022). Specifically, the study takes advantage of a textual entailment approach in which a hypothesis is formulated by entailing the aspect class or sentiment valence in a template based on the assumption that a sentence serves as a premise (Gera et al., 2022).

Suppose that given a review consisting of a sequence of sentences $S = \{s_1, s_2, \dots, s_n\}$ and a list of candidate aspects $A = \{a_1, a_2, \dots, a_n\}$, the model is required to label each pair $\{(s, a) : (s \in S, a \in A)\}$. In other words, given any s as a premise, a hypothesis is developed from a template incorporating candidate classes. This hypothesis is given as a form of a cloze prompt, such as “The subject of this sentence is { }” so that the potential classes with a probability distribution for the empty field can be predicted based on the pre-trained LLMs. To address two classification problems corresponding to each sentence, it is better to use different sub-prompts than one prompt holding two hypotheses because predicting different types of classes at the same time is challenging (Liu et al., 2023). Accordingly, as shown in Table 1, a separate sub-prompt,

Table 1
The example of textual entailment for aspect classification.

Premise (sentence)	Task	Hypothesis	
		Template	The most-likely class
“My room was one of the ‘Wellness’ rooms and offered some great features such as an air purifier and water quality and a serene setting.”	Aspect classification	The subject of this sentence is { }.	facilitates
	Sentiment classification	The sentiment of this sentence is { }.	positive

“The sentiment of this sentence is { }.”, is used, which is called “prompt decomposition” (Liu et al., 2023). As a result, each sentence generates an ‘aspect’ and ‘sentiment’ pair based on the prediction probability.

Among various LLMs for ZSTC exist, this study selected the three most reliable LLMs on the Hugging Face platform: BART (Lewis et al., 2020), DeBERTa (He et al., 2021), and RoBERTa (Liu et al., 2019). BART has auto-regressive decoders and is designed to detect fine-grained aspects from untrained and unlabeled datasets. Both DeBERTa (Decoding-enhanced BERT with Disentangled Attention) and RoBERTa (A Robustly Optimized BERT Pretraining Approach) are enhanced versions of BERT, benefiting from training larger and more diverse datasets and optimized training methods and parameters. These three models are known for their ability to predict unseen classes effectively. However, BERT was not used in this study because its architecture is based on fine-tuning with pre-annotated datasets and underperforms in the NLI task. Then, an ensemble method based on majority voting was used to obtain better outcomes, where the class that acquires two or more votes is chosen.

3.3. impact-asymmetry (IA) analysis

It is essential to exercise caution in using the predicted probability of a class as a direct indicator of the feature’s degree because the probabilities of candidate classes compete to be summed into one (Holtzman et al., 2022). Thus, this study uses the frequency of sentiment corresponding with the aspect identified within the sentences of reviews. The counted frequency is assumed to be the sentimental intensity of each aspect in reviews. Next, the study adopts a penalty-reward contrast approach in which positive and negative performances of each aspect toward overall satisfaction are separately considered to analyze the asymmetric effects of aspects (Mikulić and Prebežac, 2011). Hence, the negative sentiment associated with each aspect acts as a penalty, while the positive sentiment functions as a reward. Then, the penalty and reward scores are obtained by coefficients derived from multivariate regression analysis, shown in the equation below.

$$Satisfaction = \beta_0 + \sum_{a=1}^A (\beta_{1a}\chi_{pa} + \beta_{2a}\chi_{na}) + CV + \epsilon$$

where β_0 is the constant, and given that an aspect $a \in A$, χ_{pa} and χ_{na} are the positive and negative sentiments of the aspect a , respectively, β_{1a} and β_{2a} are the coefficients associated with those sentiments of the

aspect a , CV is a control variable, and ϵ is an error. Satisfaction is measured by user ratings.

Based on Bi et al. (2020), using the penalty and reward scores, the following equation calculates the IA score for each aspect.

$$IA_a = \frac{|\beta_{1a}| - |\beta_{2a}|}{|\beta_{1a}| + |\beta_{2a}|}$$

The IA score should range from -1 to 1 based on the formula above. When the score is close to 1 , the aspect is considered a perfect satisfier, indicating that satisfaction only exists when the performance of this aspect meets expectations. When the score approaches -1 , the aspect is identified as a perfect dissatisfier. It suggests that dissatisfaction occurs solely when the performance of this aspect fails to meet expectations. The thresholds for the IA score follow the suggestions of Lee et al. (2020). Specifically, if $IA_a > 0.8$, the aspect is classified as a delighter (an extreme satisfier); if $0.2 < IA_a \leq 0.8$, the aspect is classified as a satisfier; if $-0.2 < IA_a \leq 0.2$, the aspect is classified as a hybrid; if $-0.8 < IA_a \leq -0.2$, the aspect is classified as a dissatisfier; if $IA_a < -0.8$, the aspect is classified as a frustrator (an extreme dissatisfier).

4. Case study

4.1. Data preparation

Since the study conducts customers' ABSA using ZSTC methods, it selected online reviews in the context of COVID-19 as a case study source because they may contain previously unseen events. A customized web scraping program built on Python (version 3.9) was used to collect reviews on TripAdvisor, one of the largest social media platforms where individuals share their service experiences. Data collection was slowly processed for four months, from July 2020 to October 2020, to avoid straining the web server. The web scraper initially gathered the names of hotels from the eight leading cities. These cities were chosen according to the number of hotels they have, as reported by Statista (<https://www.statista.com/statistics/459791/number-hotel-rooms-us-cities/>). Following this, the scraper collected reviews for each of these hotels, and 'hotel name,' 'address,' 'reviewer id,' 'reviewer location,' 'rating,' 'review title,' and 'review text' items were captured and stored in the database system.

Since the burst of COVID-19 was first reported in December 2019, reviews posted after January 2020 were only considered. Among them, of particular interest are reviews explicitly mentioning the primary term, 'COVID-19.' Additionally, a word embedding technique (i.e., Word2vec) was used to identify seed words semantically related to 'COVID-19' based on the cosine similarity score. Finally, reviews containing those seed words (i.e., 'covid,' 'coronavirus,' 'virus,' 'pandemic,' 'epidemic,' 'ncov,' 'quarantine,' or 'lockdown') were selected for the analysis. A descriptive result of reviews used in this case study is shown in Table 2.

During the period, 52,707 reviews from 1819 hotels were initially collected, and 4080 reviews from 991 hotels were considered to hold

content regarding COVID-19. Orlando possesses the most significant number of hotels. However, Las Vegas stands out as having many reviews per hotel, both for all and COVID-19-related reviews. It suggests that the city's hotels are popular, and visitors are more inclined to leave reviews. Houston, while having a significant number of hotels, has a low average of reviews per hotel and a low average of COVID-19 reviews per hotel. Washington DC has the highest ratio of COVID-19-related reviews (10.7 %), indicating that a larger portion of its hotel reviews are related to COVID-19 concerns or experiences than any other city. Surprisingly, New York has the lowest number of hotels with COVID-19 reviews and the lowest ratio of COVID-19 reviews (3.7 %) despite being a major city.

4.2. Aspect-based sentiment analysis (ABSA) using zero-shot text classification (ZSTC)

Since the proposed method in this study addresses the ABSA at the sentence level, it is necessary to split a review into individual sentences. However, the creation of auxiliary sentences should be done before this splitting process to ensure that the referred named entities are not lost. Following this procedure, from a total of 4080 reviews, 45,295 sentences, each completed with auxiliary sentence construction, were identified.

Given that no pre-defined labels exist in ZSTC, this study carefully selected hospitality-specific aspect categories based on Lai and Hitchcock (2016) dimensions of satisfaction sources in hotel service. The 11 candidate aspects (facilities, staff, location, food, price, cleanliness, design features, technologies, entertainment, safety, and general) were specifically chosen to

Table 3

The result of robust regression analysis and impact-asymmetry analysis.

Variable	Estimate	Aspect	IA score	Factor
facilities_pos	1.409*** (0.122)	Facilities	-0.268	dissatisfiers
facilities_neg	-2.442*** (0.110)			
staff_pos	2.142*** (0.146)	Staff	-0.147	hybrids
staff_neg	-2.881*** (0.200)			
location_pos	1.423*** (0.137)	Location	-0.116	hybrids
location_neg	-1.797*** (0.163)			
food_pos	0.430* (0.197)	Food	-0.319	dissatisfiers
food_neg	-0.833*** (0.242)			
price_pos	1.386*** (0.359)	Price	-0.345	dissatisfiers
price_neg	-2.846*** (0.211)			
cleanliness_pos	2.414*** (0.266)	Cleanliness	-0.076	hybrids
cleanliness_neg	-2.810*** (0.305)			
design_pos	1.200*** (0.242)	Design	0.878	delighters
design_neg	-0.078 (0.516)			
technologies_pos	-0.226 (0.883)	Technologies	-0.737	dissatisfiers
technologies_neg	-1.495* (0.804)			
entertainment_pos	0.793*** (0.304)	Entertainment	-0.302	dissatisfiers
entertainment_neg	-1.481** (0.617)			
safety_pos	2.470*** (0.284)	Safety	0.444	satisfiers
safety_neg	-0.952*** (0.268)			

Notes: N = 4080; *p < 0.1; **p < 0.05; ***p < 0.01; The number of sentences is a control variable; IA score = Impact-asymmetry score.

Table 2

The descriptive summary of the focal data.

Location	All reviews			COVID-19 related reviews			
	No of hotels	No of reviews (1)	Review per hotel	No of hotels	No of reviews (2)	Review per hotel	Ratio (2) / (1)
Orlando	327	11200	34.3	226	966	4.3	8.6 %
LA	320	4693	14.7	138	456	3.3	9.7 %
Houston	320	3162	9.9	115	210	1.8	6.6 %
Las Vegas	201	16170	80.4	132	1190	9.0	7.4 %
New York	181	6424	35.5	84	236	2.8	3.7 %
Chicago	171	4394	25.7	114	350	3.1	8.0 %
Dallas	151	2674	17.7	84	246	2.9	9.2 %
Washington DC	148	3990	27.0	98	426	4.3	10.7 %
Total	1819	52707	29.0	991	4080	4.1	7.7 %

capture the key dimensions of hotel experiences. As described in Table 3, a hypothesis template using three LLMs (i.e., BART, DeBERTa, and RoBERTa) predicted the most-likely class of an aspect with distributed probabilities. To maximize the accuracy, this study used models having a large size of parameters, that is, ‘facebook/bart-large-mnli’ for BART, ‘MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli’ for DeBERTa, and ‘roberta-large-mnli’ for RoBERTa, respectively on the Hugging Face platform. All these ZSTC procedures were conducted using PyTorch v2.0 (<https://pytorch.org/>). Then, the predicted class that received the majority vote from the three LLMs was designated as the final class. In case all three models predicted different classes, one with the highest probability score was selected. Regarding the classification of sentiment valence, one of three sentiment labels (i.e., positive, negative, and neutral) was allocated for each sentence in the same way as the majority voting of aspect classification.

In the subsequent step, as demonstrated in Fig. 2, this study converted the aspect-sentiment pair set of each sentence into quantitative values at the review level based on their frequency of occurrence. Since each sentence within a review contains a unique aspect-sentiment pair set and a review is composed of multiple sentences, the study restructured the sentence records into a wide format, including the aspect-sentiment pairs as independent variables. The count of each pair set was tallied whenever it appeared, augmenting the corresponding variable’s value and thereby intensifying the sentiment. Through this process, the study quantified the overall sentiment intensity for all aspects within a review.

This study generated a correlation map to ensure that LLMs judge aspects’ sentimental polarities appropriately. As shown in Fig. 3, all the variables of positive sentiments are positively related to ratings (colored in blue). In contrast, all the variables of negative sentiments are negatively related to ratings (colored in red), which supports an assumption that the result of ZSTC is reliable. Moreover, the number of sentences in a review was added as a control variable because the appearance frequency of aspects and sentiments is impacted by the number of sentences used in a review. The correlation map reveals that the number of sentences is negatively related to ratings, which is consistent with previous studies demonstrating that the length of a review is negatively related to ratings (Ghasemaghaei et al., 2018).

4.3. impact-asymmetry (IA) analysis

In this study’s multivariate regression analysis, a breach of multivariate normality (HZ statistic = 21.42, $p < .001$) was observed, as indicated by the Henze–Zirkler (HZ) test (Henze and Zirkler, 1990). Moreover, the studentized Breusch–Pagan test (Koenker, 1981) showed heteroscedasticity in the OLS regression model (BP statistic (4080, 21) = 453.19, $p < .001$). To address the violation of these assumptions, this study conducted a robust regression analysis (Huber and Ronchetti, 2009). As displayed in Table 3, the results show that all variables are statistically significant except for ‘design_neg’ and ‘technology_pos.’ Further, this study calculated the IA scores of aspects using the formula described above. Based on the established criteria, five aspects (e.g., facilities, food, price, technologies, and entertainment) were categorized as dissatisfiers, three aspects (e.g., staff, location, and cleanliness) were identified as hybrids, and design and safety were classified as delighters and satisfiers, respectively.

Then, the study designed the impact level-asymmetry map, as shown in Fig. 4. This approach facilitates the identification of more or less relevant aspects of customer satisfaction and improvement priorities (Mikulić and Prebežac, 2008). For instance, ‘technologies’ is a dissatisfier that must be adequately provided to prevent dissatisfaction. However, its strength of impact on customer satisfaction is feeble. In contrast, ‘facilities’ is also dissatisfier, but given its medium impact, it should have higher priority. Delighters and satisfiers are positioned on the upper side, and ‘design’ and ‘safety’ fall into this category, respectively. They currently exceed customer expectations but have a low or moderate impact, implying a good performance but less necessity for immediate improvement than other aspects. For hybrids, all the aspects have medium to high impact levels with moderate asymmetry, contributing to both satisfaction and dissatisfaction.

4.4. Robustness checks

4.4.1. Robustness check with human annotations

In a scenario where ZSTC does not presume any annotated dataset, model performance evaluation is not naturally available, and no ground-

Hotel level	Review level	Sentence level	Aspect level	Aspect1_ positive	Aspect1_ negative	Aspect2_ positive	...	Aspect12_ negative
Hotel 1	Review 1	Sentence 1	[a2, pos]	-----				
		Sentence 2	[a1, pos]	-----→ 2	0	1	...	0
		Sentence 3	[a1, pos]	-----↑				
		⋮	⋮					
	Review 2	Sentence 1	[a12, neg]					
		Sentence 2	[a1, pos]	1	0	1	...	3
		Sentence 3	[a2, pos]					
		⋮	⋮					
	Review 3			0	2	0	...	1
	⋮							
	Review N							
Hotel 2	Review 1							
⋮	⋮							

Fig. 2. The restructured form of the dataset. a1 and a2 denote the first and the second aspects.

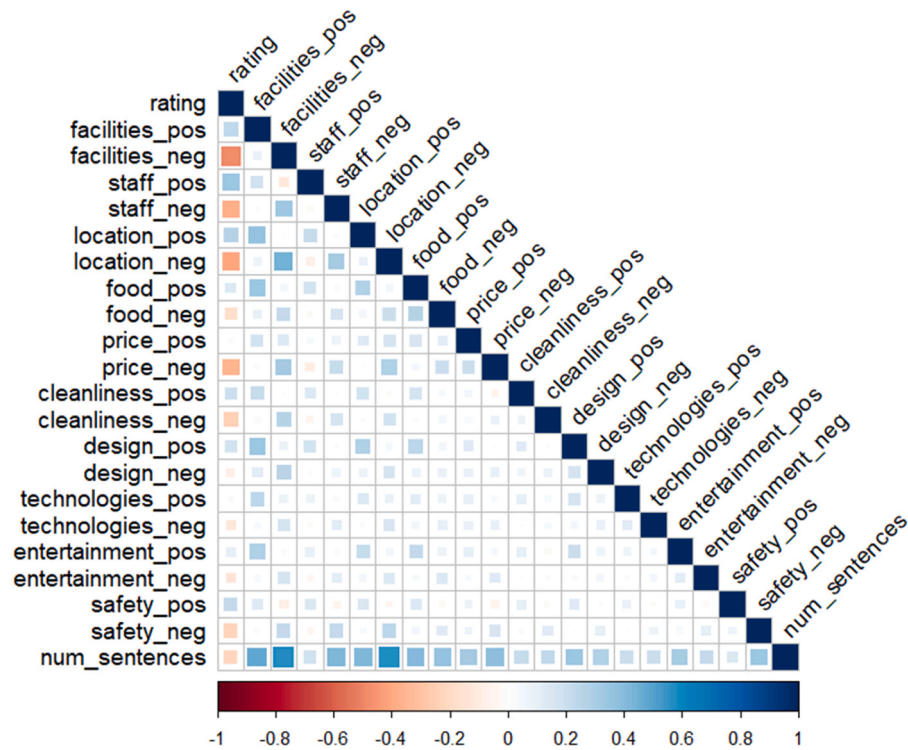


Fig. 3. The correlation map.

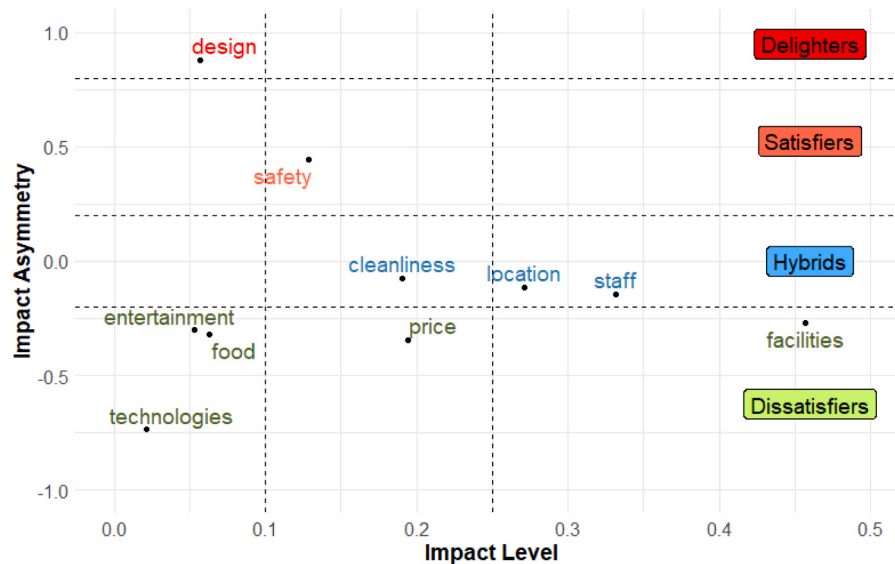


Fig. 4. The impact level-asymmetry map.

truth classification can be predicted with proven accuracy (Basile et al., 2021). This study, therefore, had the dataset manually annotated to compare labels with the results predicted by the ZSTC for the robustness check. First, the researcher randomly sampled 1 % of 45,295 sentences using a stratified sampling method because the numbers of predicted aspects were not equally balanced. Thus, 453 sentences were distributed to three graduate students, who then annotated their aspects and sentiments. If any of two or more manual annotations were equal, it was finally confirmed as the final label. If none of the three was equal (i.e., 67 cases in the aspect and 20 cases in the sentiment), the researcher selected one among the candidates.

The reliability of ZSTC results was assessed by the comparison of

Cohen's kappa coefficients (Cohen, 1960) with manual annotations across 11 aspect categories and three sentiment categories. Cohen's kappa is known as a robust measure for the degree to which shows the performance of classification (Kirilenko et al., 2018), and management and marketing studies have employed it to evaluate the reliability of rater or annotation results (e.g., Hulland et al., 2018; Mirtalaie and Hussain, 2020; Wu et al., 2020). As shown in Table 4, aggregated values from multiple human annotators yield higher Cohen's kappa coefficients (0.40 for the aspect and 0.64 for the sentiment, emphasized in bold in the table) with ZSTC than those from any other individual annotators, suggesting that the ZSTC is efficient because it means that ZSTC captures a more collective human judgment. Furthermore, the findings indicate

Table 4

Cohen's kappa and consistency between human annotations and ZSTC.

Annotation	Cohen's κ		Consistency	
	Aspect	Sentiment	Aspect	Sentiment
HA1	0.27 [0.22, 0.32]	0.48 [0.41, 0.55]	37.97 %	70.64 %
HA2	0.35 [0.29, 0.40]	0.62 [0.56, 0.67]	45.47 %	75.94 %
HA3	0.36 [0.30, 0.41]	0.50 [0.45, 0.56]	44.81 %	67.99 %
HA-all	0.40 [0.34, 0.45]	0.64 [0.58, 0.69]	49.45 %	77.92 %

Notes: N = 453. HA1, HA2, and HA3 denote human annotators who manually labeled classes. HA-all denotes the combination of all human annotators. The numbers in a bracket are the lowest and the highest coefficients in the 95 % confidence interval.

that the reliability of the aspect is nearly moderate ($\kappa > 0.4$), while sentiment demonstrates substantial ($\kappa > 0.6$) inter-reliability (Landis and Koch, 1977). The consistency between ZSTC and human-annotated results has the same pattern as Cohen's kappa, yielding the highest consistency for the aggregated value.

4.4.2. Robustness check with training dataset

To further validate the proposed method and address the need for comprehensive model comparisons, this study applied the method to the standard training dataset (SemEval2014) available on Kaggle (<https://www.kaggle.com>) and compared it with traditional ABSA approaches. Although this dataset is acquired and labeled in a restaurant domain, there is no comparable training dataset for the hotel domain, making it challenging to apply traditional machine-learning approaches to ABSA for hotel reviews. This study chose the five most frequent aspect categories from the SemEval2014 training dataset, excluding a miscellaneous category, because the dataset entails almost 1300 labels, with duplicates like 'price' and 'prices.' After this selection, a total of 867 labeled records with aspect categories such as 'food,' 'service,' 'ambience,' 'price,' and 'place' were used as the secondary source for robustness validation.

While BERT was designed initially for pre-training and is not ideally suited for NLI, it was included as a baseline model to evaluate ZSTC in a scenario in which no-domain specific annotated data is available. Additionally, this study included a fine-tuned BERT model as a reference point to demonstrate the performance ceiling achievable with supervised learning approaches that have access to labeled training data. To provide more comprehensive comparison with traditional approaches, the study included VADER, a widely-used lexicon-based SA technique that represents conventional approaches in the field. However, as VADER is primarily a SA tool without built-in aspect classification capabilities, it could only be evaluated on sentiment classification tasks. The performance of the proposed model was assessed for both aspect and sentiment classification, with comprehensive evaluation metrics including accuracy, precision, recall, and F1-score, as presented in Table 5.

As expected, BERT without fine-tuning performed poorly across both tasks, demonstrating the necessity of alternative approaches. Remarkably, the proposed ZSTC ensemble model achieved performance comparable to the supervised fine-tuned BERT model in aspect classification accuracy (0.6471 vs. 0.6724) and outperformed it in sentiment

classification accuracy (0.8166 vs. 0.7701), despite relying on no labeled training data. The F1 scores reveal a more nuanced performance profile. The method demonstrates competitive result in aspect classification (0.5488 vs 0.6746) and sentiment classification (0.6307 vs 0.7677). It should be noted that the fine-tuned BERT is used solely as a reference point and is not directly comparable to the ZSTC ensemble method proposed in this study, since it requires extensive labeled data for training, which is precisely the limitation the study seeks to overcome.

Importantly, when compared to traditional approaches commonly used by researchers, the method shows substantial improvements. Relative to VADER, it illustrates 23.8 % increase in sentiment classification accuracy (0.8166 vs. 0.6597) and 19.9 % gain in F1 score (0.6307 vs 0.5261), while also providing aspect classification capabilities that conventional lexicon-based tools entirely lack.

5. Discussion

Natural language processing has become ubiquitous in social science journals. The recent advancement of LLMs in computer science has shown great potential but less affected hospitality and tourism management literature. As shown in Fig. 1, this study proposes an overarching framework applying ZSTC to ABSA, which leads to IA analysis to enrich an understanding of aspects' linear or nonlinear effects on customer satisfaction. The entire process aligns with how to gain insights into the specific aspects and their roles in customer satisfaction by presenting customer-generated texts into pre-trained LLMs without a pre-labeled dataset. A case study adapting proposed methods produces effective results, which evoke subsequent theoretical and practical discussions.

5.1. Theoretical implications

With the rapid increase of textual content on social media, how to leverage these online texts encompassing a range of customer perceptions and attitudes on products and services has become very important. Specifically, rather than referring to aggregated evaluation, delving into positive or negative sentiments of various aspects can capture multiple dimensions. In this regard, ABSA has recently attracted academic attention since analyzing sentiment polarity at a document level may lead to an insufficient understanding. However, the use of ABSA has been limited in management literature. It requires solutions to address issues to be effective: identifying aspects and detecting sentiment valence matched with the identified aspects, particularly when labeled data are not available

This study proposes state-of-the-art methods based on the ZSTC that address these issues, with significant implications for hospitality management theories. First, by using pre-trained LLMs, this approach classifies aspects at a sentence level and determines sentiment polarity for the identified aspect without requiring hospitality-specific training data. This methodological innovation directly addresses a persistent barrier to advanced customer analytics in the hospitality research, thus enabling more sophisticated investigation of service quality dimensions and their impact on satisfaction. This approach effectively overcomes the ABSA's

Table 5

The performance comparison between BERT, VADER, and the proposed model using ZSTC.

Model	Aspect				Sentiment			
	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1
BERT (a baseline)	0.0346	0.0709	0.0297	0.0371	0.1373	0.2045	0.2697	0.1374
BERT (fine-tuned)	0.6724	0.6802	0.6724	0.6746	0.7701	0.7666	0.7701	0.7677
VADER SA	N/A	N/A	N/A	N/A	0.6597	0.5625	0.5311	0.5261
Proposed ensemble model	0.6471	0.5728	0.5695	0.5488	0.8166	0.6655	0.6345	0.6307

Note: The fine-tuned BERT model was trained on 80 % of the dataset (training set), and its performance was evaluated on the remaining 20 % (validation set). It should be noted that the fine-tuned BERT model is feasible only when a pre-trained dataset with same category labels exists.

challenge of recognizing a sentiment polarity that aligns with the specified aspect.

Second, it does not need to rely on lexicons as an outward reference, even when labeled data is unavailable. Third, since it uses LLMs that employ contextual embedding, the result of ZSTC can recognize the distinct meanings of the polysemous words, ensuring higher accuracy. This method applies to various domains and circumstances because it allows researchers to establish any classes of their interests. Specifically, it can investigate conceptually innovative features for emerging phenomena like COVID-19, as done in the case study presented here. Furthermore, it can interpret complicated and multidimensional facets of concepts implied in a document, which has excellent potential for extending social science research. Furthermore, to enhance trustworthiness, this study conducts two robustness checks, which compare the result generated by an ensemble model of three LLMs with human annotations (the first robustness check), and with pre-classified aspects and sentiments (the second robustness check).

Although this study primarily focuses on proposing an innovative methodology for ABSA, its theoretical contribution to the literature is substantial. As noted by Mehraliyev et al. (2022), sentiment holds the potential to advance beyond regression analysis. Particularly in ABSA research, while Li et al. (2023) investigated the importance of aspects for restaurant survival in their ABSA research, few studies have examined the dynamic role of each aspect in shaping customer satisfaction. This study applied the proposed method within the framework of the Kano model and incorporates impact-asymmetry analysis to explore asymmetric relationships with satisfaction. As aspects can be considered as factors in textual data, the capability of identifying sentiment levels associated with an individual aspect enables researchers to test hypotheses through identified aspects and sentiments across vast volumes of texts.

The proposed methodological framework represents a fundamentally different paradigm from generative LLM approaches for several practical and technical reasons. Processing large-scale hospitality review datasets (such as 45,295 sentences in this paper) requires efficient batch processing capabilities that our ensemble ZSTC approach provides through vectorized operations. Generative LLMs operate through individual prompt-response interactions requiring separate API calls for each classification task. They create cost inefficiencies and process bottlenecks unsuitable for research-scale analysis. Furthermore, the integrated analytical framework that combines ensemble ZSTC with IA analysis addresses the persistent challenge of conducting ABSA without domain-specific annotated datasets, while establishing a replicable methodology that transcends individual model architectures.

5.2. Practical implications

At a document level, sentiment analysis is insufficient to understand customers comprehensively. Although a firm can gain some insights into the aggregated results of the analysis, it cannot often find where to go with those results. If it captures customers' sentiment polarity at an aspect level, it can have a better chance to satisfy their expectations. Hence, ABSA is critical to having a competitive advantage from a firm's perspective. ABSA is valuable for in-depth understanding even when the customers' quantitative score (e.g., a rating) is unavailable. However, as mentioned above, ABSA has some disadvantages for ease of use. Most companies, for example, do not have the resources to adapt up-to-date ABSA methods that generally require a large dataset.

The proposed methods in this study are easy to follow and provide alternative solutions. By taking advantage of pre-trained LLMs, ZSTC does not need to build any complicated model. Besides, it does not need human annotation or data training work, which are burdensome for most companies. Therefore, it can flexibly ensure the sentiment polarity for any new aspect and reflect a change in business environments, which previously required constructing and training a new dataset. In this regard, the proposed methods contribute to new product development

where there is no information on customer preferences for the product. Since ZSTC can extract sentiments of attributes in virtually any product similar to the planned product, those identified sentiments corresponding to the targeted attributes can be mapped into the components as references for the new product.

Most studies concentrate on the aspects of ameliorating customer satisfaction. However, those aspects are not linearly related to customer satisfaction. Dissatisfiers are only related to dissatisfaction when customers' expectations are not met, while satisfiers are only related to satisfaction when customers' expectations are met. Despite the successful identification of aspects using ABSA, finding out how they are related to an accumulated satisfaction is demanding (Chang et al., 2022). The case study based on the proposed methods shows how to classify identified aspects into dissatisfiers, satisfiers, and hybrids. Furthermore, a map showing the impact level and asymmetry highlights the priorities of different aspects and can aid in enhancing companies' decision-making processes.

Thus, ABSA using ZSTC enables companies to understand customer perceptions without intensive human labor. Even the voices of customer calls can be transformed into a textual format so that the proposed methods in this study can be employed. Moreover, firms that hesitate to invest in the analysis of customer sentiments at a granular level do not need to suffer from expenses since the methods are flexible in future changes or class expansions. It assists companies in enhancing their decision-making processes, thereby maintaining a competitive edge.

5.3. Limitations and future research

This study proposed state-of-the-art methods, and a case study demonstrated their feasibility. However, it has limitations that may guide future research directions. First, NLI based on LLM is subject to potential biases rooted in the collection of datasets on which the model is trained. Even models trained on vast datasets contain inherent biases related to factors such as race, gender, and nationality. Additionally, each model draws from different data sources, meaning that the performance of LLMs can vary depending on the application domain. This study employed an ensemble model that integrates three advanced models to mitigate potential biases associated with any individual model and achieve more consistent results. As this is a critical research area in computational linguistics, researchers in hospitality and tourism should closely monitor the progress of LLM techniques. In this regard, the innovation of LLMs is ongoing, which will further enhance the overall quality of NLI.

Second, one of the advantages of ZSTC is that the researcher can freely define classes in advance. However, as the number of classes grows, classification accuracy will decrease. Unfortunately, the optimized number of classes is unknown. As the optimized number of topics is unknown in topic modeling, it is an overall problem with unsupervised learning. However, there is a need to continue research on improving and finding an adequate number of classes in line with the development of NLP using LLMs. Furthermore, from a prompt engineering perspective, future research could adopt the development of more effective prompts or the use of ensemble prompt methods, which is an active area of discussion in current NLP research.

Third, in the case study, this study only focused on reviews related to COVID-19, excluding other data, such as reviews before 2020, for the comparison as it was beyond the scope of this study. It will be meaningful to compare changes in the sentiment polarity in aspects and their impact levels over different periods in the future. Moreover, the proposed methods can be applied to other contexts where datasets with annotation do not exist. Integrating the ZSTC and the Kano model can transform qualitative evaluations of aspects into quantitative features for empirical research. Using this study as an example, future research can expand upon these findings by comparing results based on different conditions, such as the grades or locations of hotels. Researchers could apply the proposed methods to different contexts of their interests.

6. Conclusion

In the hospitality and tourism context, ABSA presents particular value as service experiences are inherently multidimensional. Traditional hospitality research has relied on structured surveys to capture these dimensions, whereas user-generated content can offer richer, unprompted insights. However, hospitality researchers face unique challenges in applying ABSA: (1) the diverse and evolving nature of service attributes that vary across property types and service levels, (2) the lack of hospitality-specific labeled datasets for training supervised models, and (3) the need to connect sentiment analysis with service management frameworks that can guide operational improvements.

This study addresses these hospitality-focused challenges by proposing an innovative methodology that integrates ABSA using ZSTC with impact-asymmetry analysis based on the Kano model. This approach eliminates the need for domain-specific training data or extensive annotation processes, while directly identifying aspect categories based on pre-defined targets. This methodological innovation transforms qualitative customer feedback into quantitative variables that support theory-based decision-making.

The theoretical contribution lies in advancing our understanding of how different service aspects asymmetrically influence customer satisfaction in hospitality settings, filling a significant gap in ABSA research using LLMs within hospitality contexts. The methodological framework provides empirical support for non-linear relationships between service dimensions and overall satisfaction, which contributes to theory building in hospitality management. For practical application, this framework enables hotel managers to strategically interpret which service elements require immediate attention to prevent dissatisfaction (e.g., facilities and price in the case study) versus which elements drive exceptional satisfaction when executed well (e.g., safety and design features). By classifying aspects as satisfiers, dissatisfiers, or hybrids, managers can prioritize service improvements effectively. This nuanced understanding is particularly valuable in resource-constrained environments.

CRedit authorship contribution statement

Wooseok Kwon: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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